

(TR-104) PYTHON WITH AI/ML

Training Day 21 Report

Date: 9 February 2026

Topic Covered: Automation, Data Pipeline Development & Introduction to NLP Concepts

The twenty-first day of training focused on completing advanced automation and data pipeline projects while also introducing foundational concepts of Natural Language Processing (NLP). The day emphasized building reliable, production-style Python systems and understanding how machines begin to process and interpret text data.

- Completed automation-based project involving file system operations.
- Implemented file monitoring system for detecting new files.
- Applied validation and processing logic on incoming files.
- Moved processed files to output directories with structured handling.
- Ensured proper error handling and logging throughout the workflow.
- Developed an end-to-end data pipeline integrating data fetching, processing, and storage.
- Implemented asynchronous data handling for efficient execution.
- Added retry mechanisms and centralized logging for reliability.
- Used configuration-based approach to avoid hardcoding.
- Gained exposure to os, shutil, and file handling automation concepts.
- Explored asynchronous programming and logging in real-world workflows.
- Introduced to NLP fundamentals including its benefits, approaches, key tasks, and challenges.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/PROJECT-9-End-to-End-Data-Pipeline-CAPSTONE->
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/the-journey-of-data-how-end-to-end-pipelines-transform-insights-32cd48fbf819>